CLOUD SECURITY SPECIALIST

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LAB:CONFIGURING AND SECURING ACR AND ACKS

1.1 INTRODUCTION

I started the lab with a required lab set up.I clicked on ctrl alt and delete to bring up the login page,signed in as admin and then downloaded the lab files.

1.2 LAB 04:CONFIGURING AND SECURING ACR AND ACKS

LAB SCENARIO

I was asked to deploy a proof of concept with Azure Container Registry and Azure Kubernetes Service.The proof of concept would demonstrate:Using Dockerfile to build an image,Using Azure Container Registry to store images,Configuring an Azure Kubernetes Service and Securing and accessing container applications both internally and externally.

EXERCISE 1:CONFIGURING AND SECURING ACR AND ACKS

In this exercise I created an Azure Container Registry,created a Dockerfile, build a container and pushed it to Azure Container Registry,created an Azure Kubernetes Service cluster,Granted the AKS cluster permissions to access the ACR,Deployed an external service to AKS,Verified the you can access an external AKS-hosted service,Deployed an internal service to AKS and Verified the you can access an internal AKS-hosted service.

TASK 1

In this task, Icreated a resource group for the lab, an Azure Container Registry.

Step 1

I signed into the azure portal with the credentials provided by the lab host

Step 2

In the Azure portal, I opened the Cloud Shell by clicking the first icon in the top right of the Azure Portal. When prompted, clicked Bash and Created storage.

Step 3

I ensured Bash was selected

Step 4

In the Bash session within the Cloud Shell pane, I ran the command to create a new resource group for the lab.

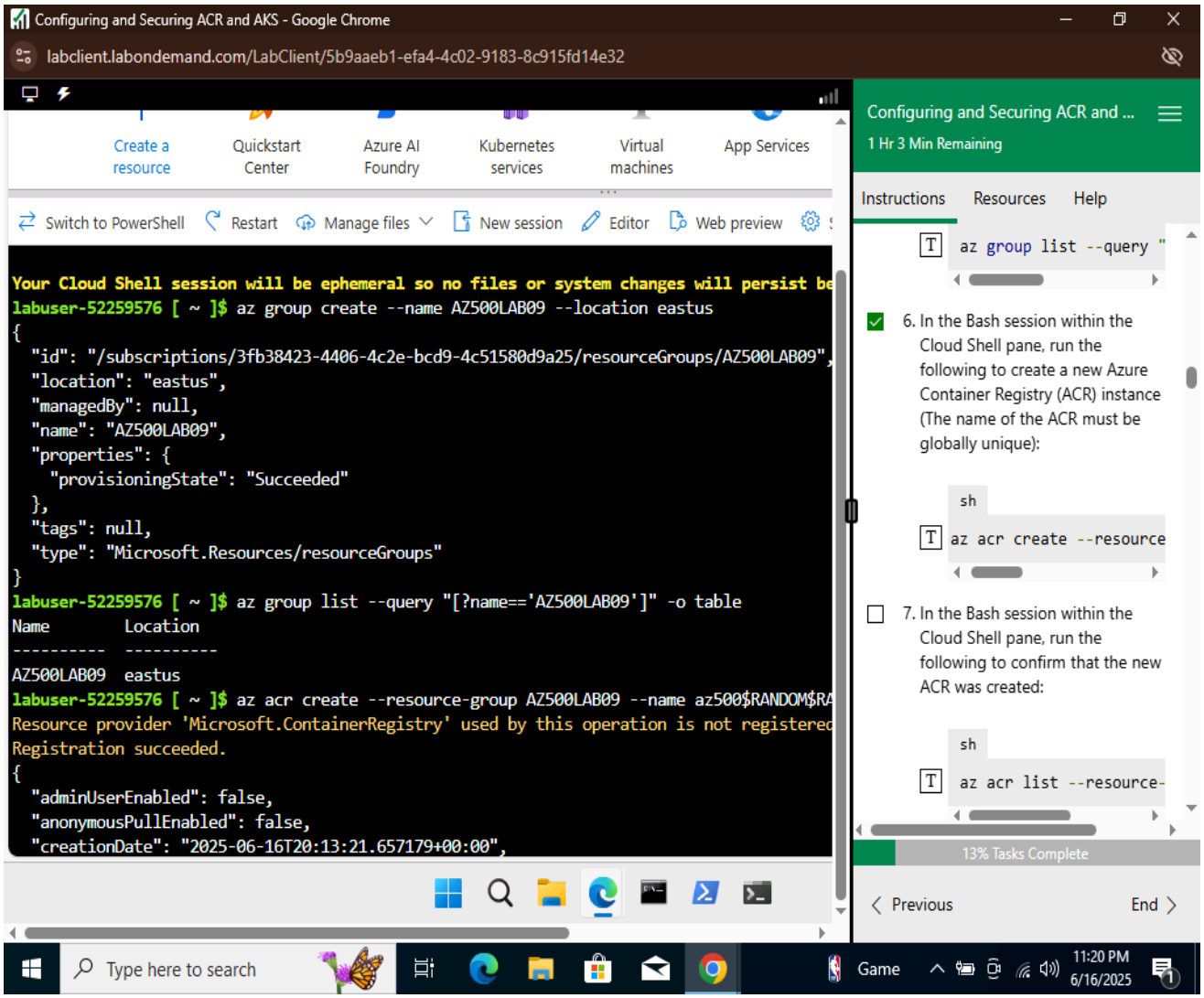
Step 5

I then ran the command to verify the resource group was created.

Step 6

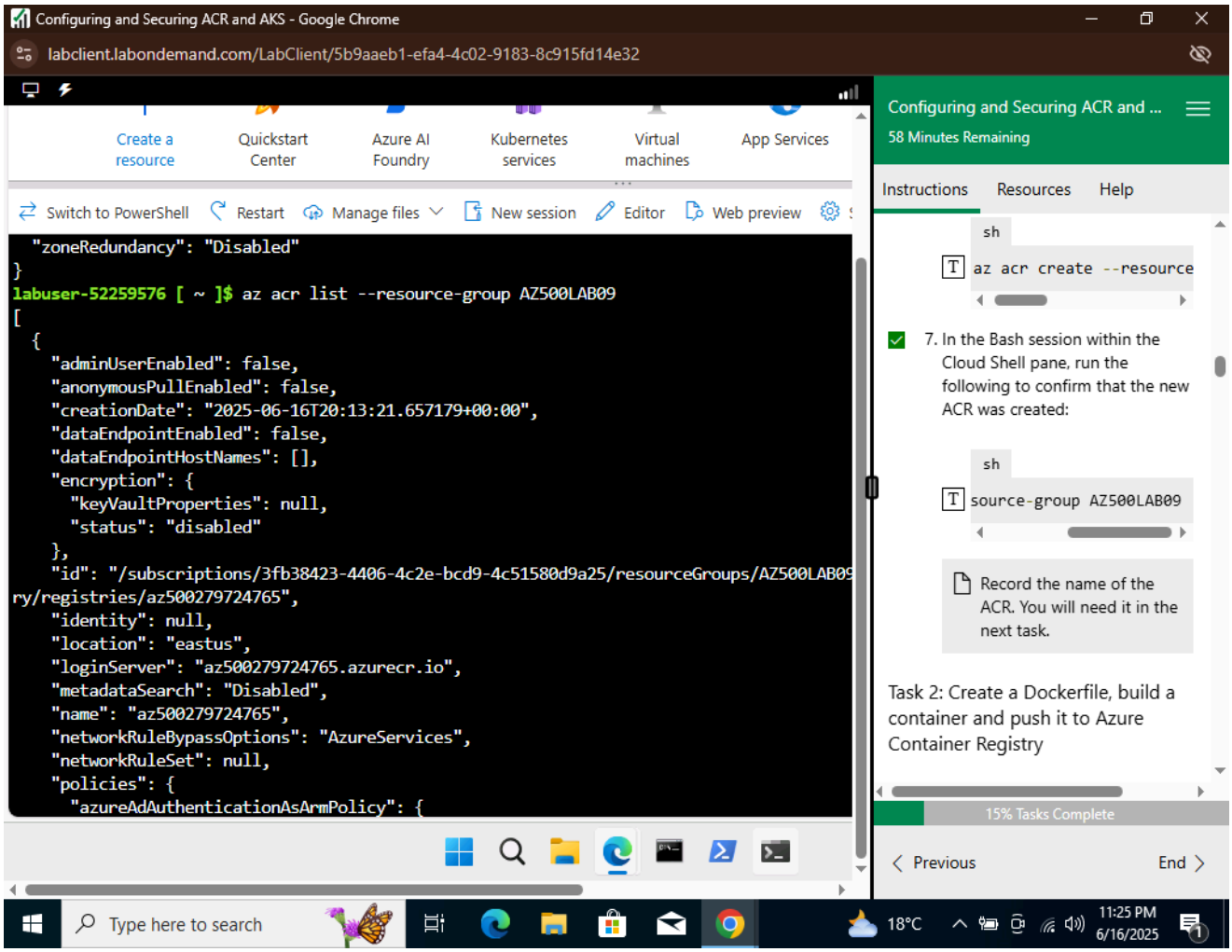
In the Bash session, I ran the command to create a new Azure Container Registry (ACR) instance.

Below is my screenshot for steps 4,5 and 6



Step 7

In the Bash session within the Cloud Shell pane, I ran the command to confirm that the new ACR was created.Below is my screenshot



TASK 2

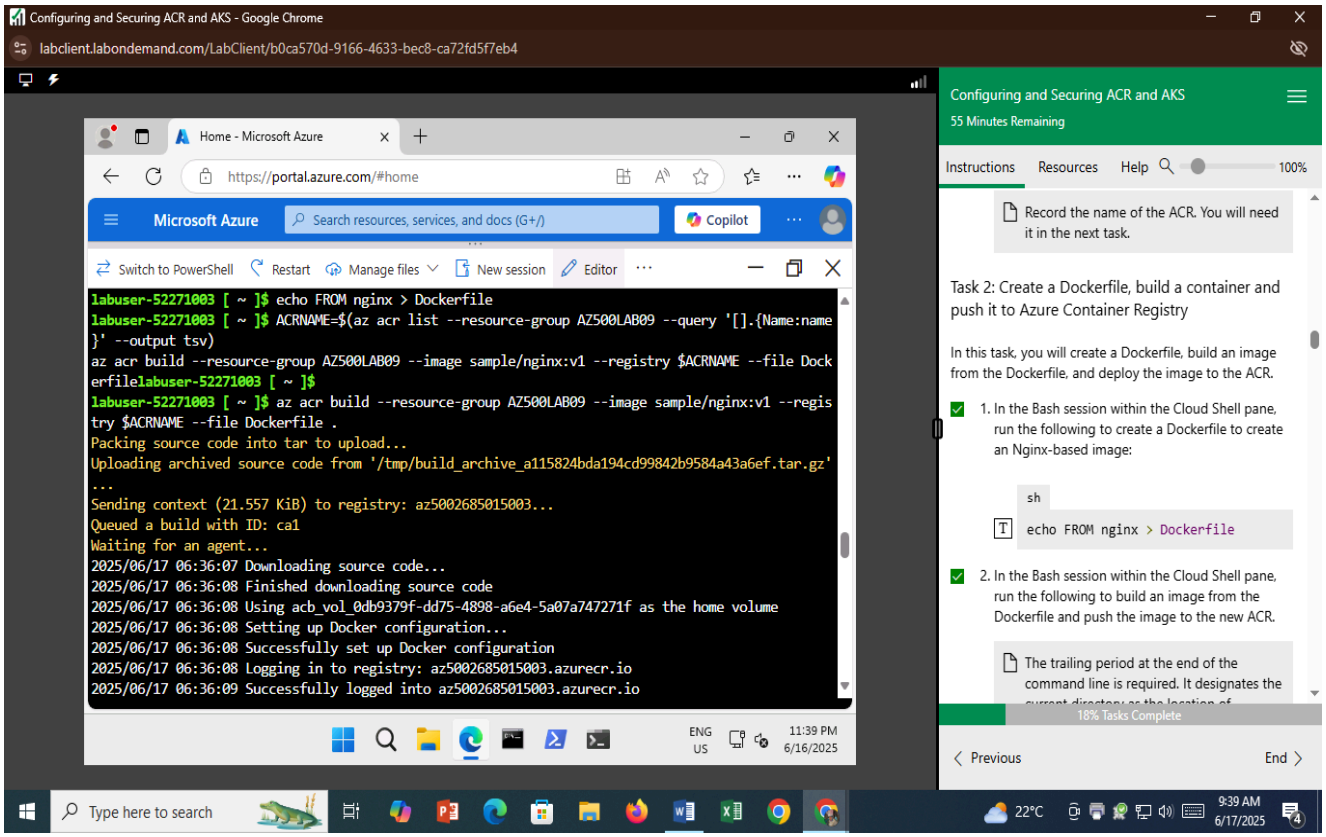
In this task, I created a Dockerfile, built an image from the Dockerfile, and deployed the image to the ACR.

Step 1

In the Bash session within the Cloud Shell pane, I ran the command to create a Dockerfile to create an Nginx-based image

Step 2

I then ran the command provided to build an image from the Dockerfile and push the image to the new ACR. Below is my screenshot for step 1 and 2.



Step 3

I closed the Cloud Shell pane.

Step 4

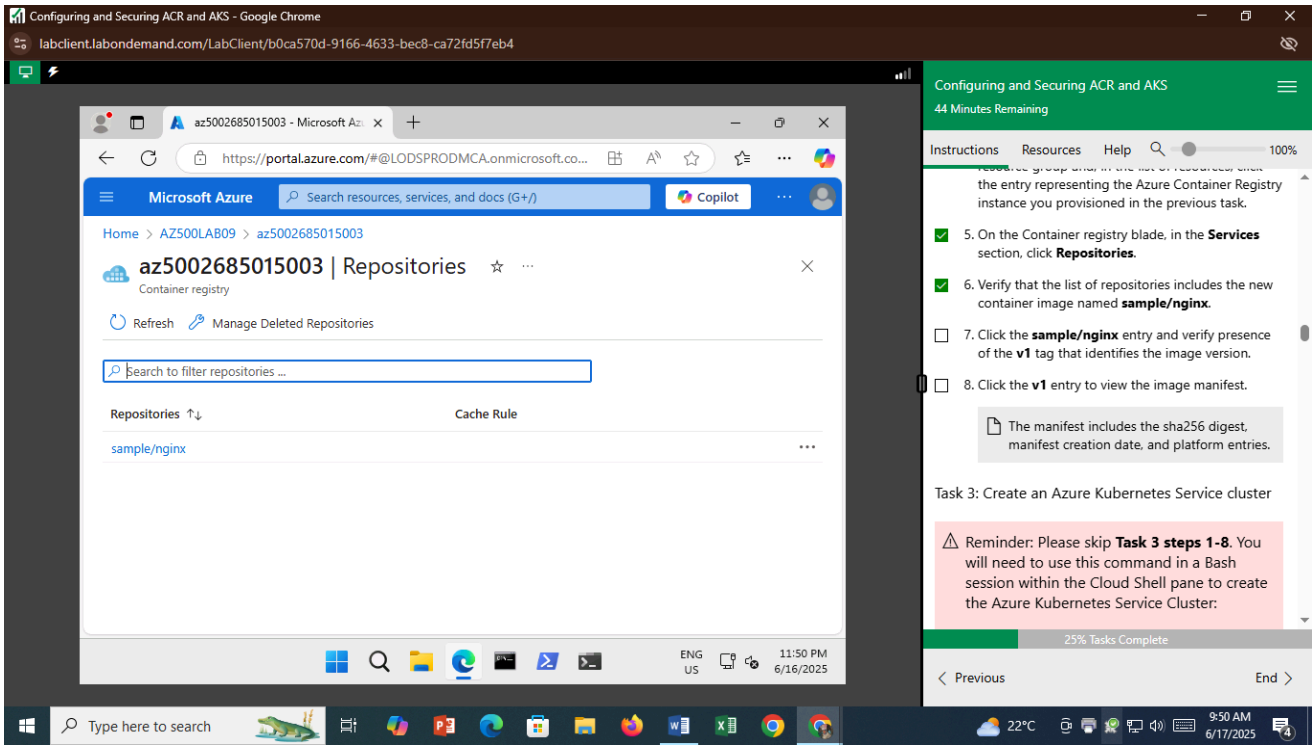
In the Azure portal, I navigated to the AZ500LAB09 resource group and, in the list of resources, clicked the entry representing the Azure Container Registry instance I provisioned in the previous task.

Step 5

On the Container registry blade, in the Services section, I clicked on Repositories.

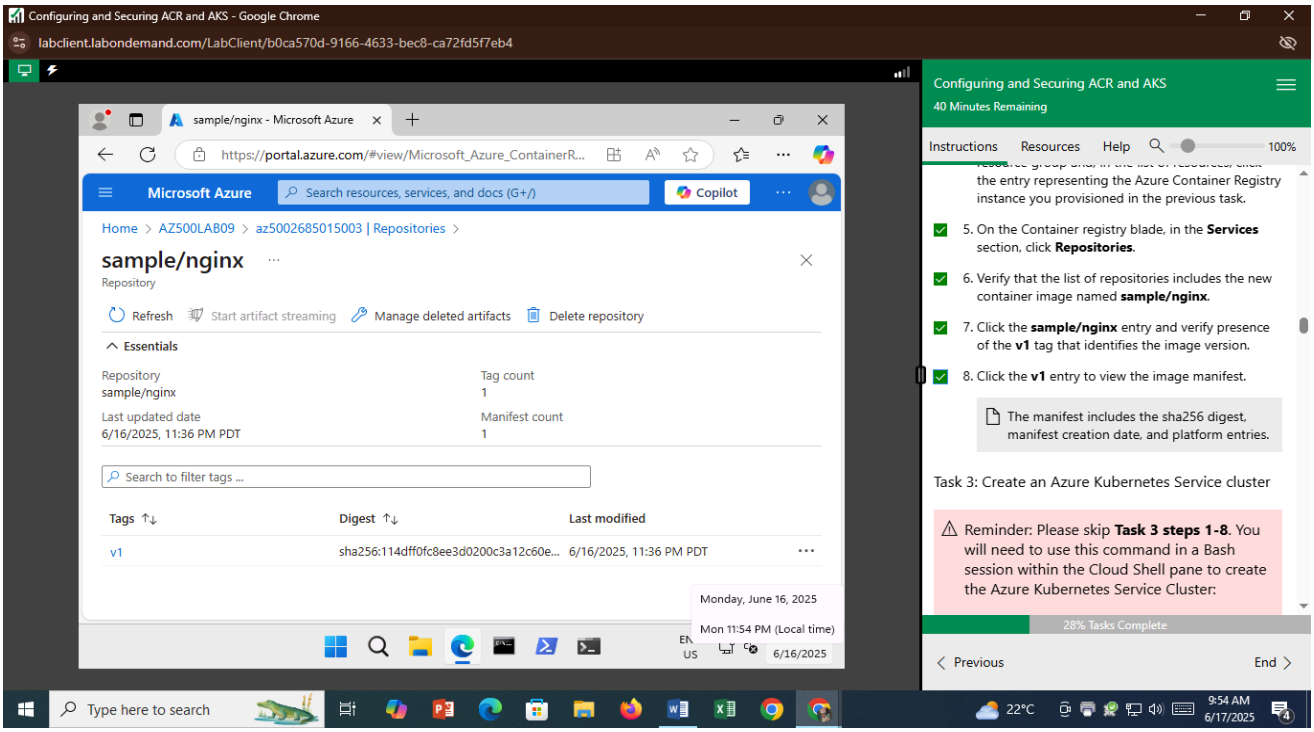
Step 6

I verified that the list of repositories included the new container image named sample/nginx.Below is my screenshot



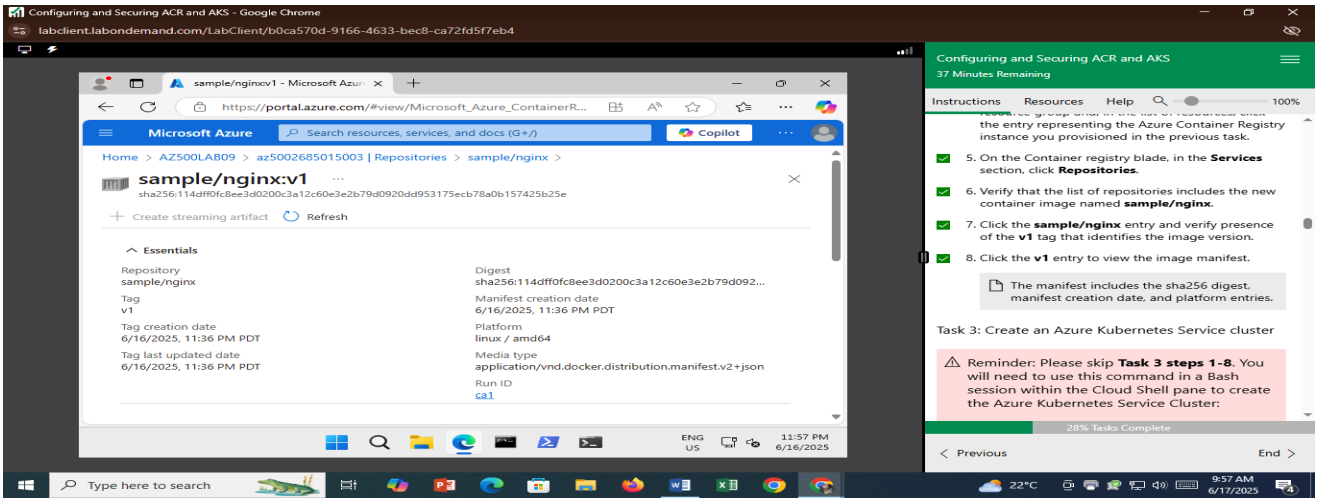
Step 7

I clicked the sample/nginx entry and verified presence of the v1 tag that identifies the image version.Here is my screenshot



Step 8

I clicked the v1 entry to view the image manifest.The manifest included the sha256 digest, manifest creation date, and platform entries.Here is my screenshot

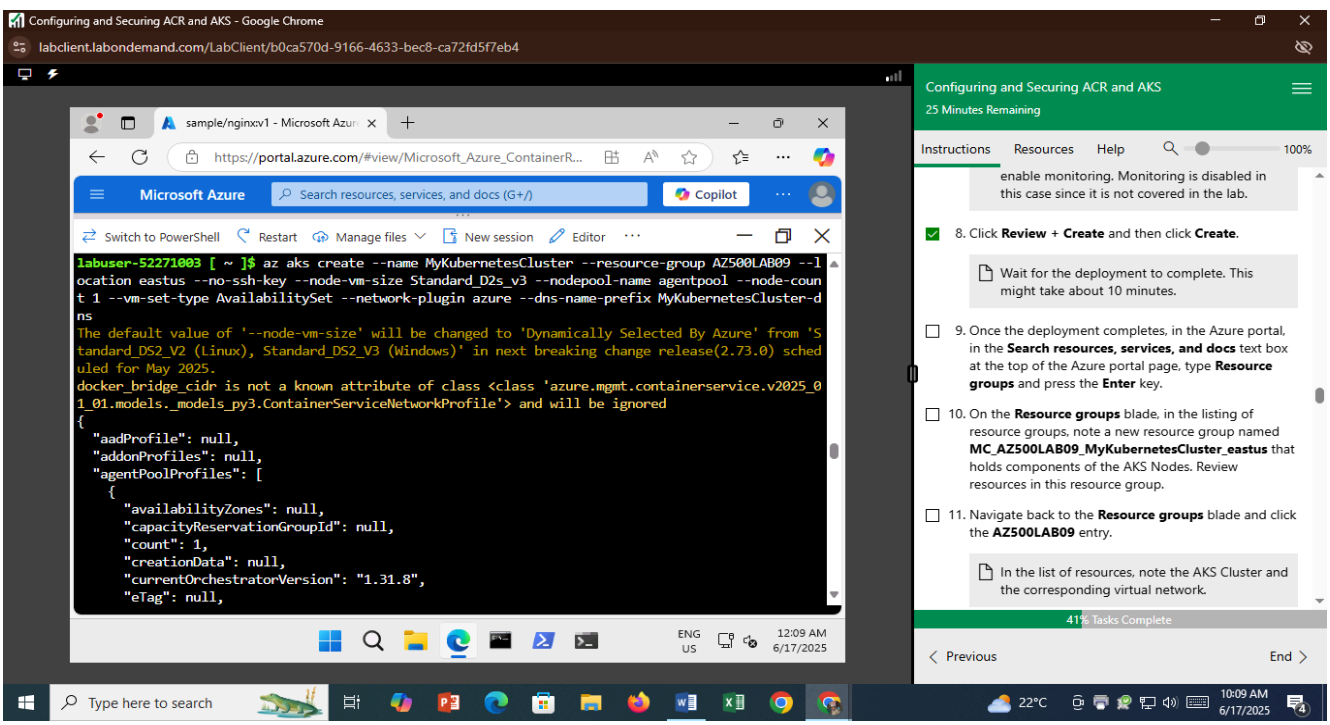


TASK 3

In this task,I created an Azure Kubernetes service and reviewed the deployed resources.

Step 1

I skipped the steps 1-8.because I used the command provided in a Bash session within the Cloud Shell pane to create the Azure Kubernetes Service Cluster.Below is my screenshot

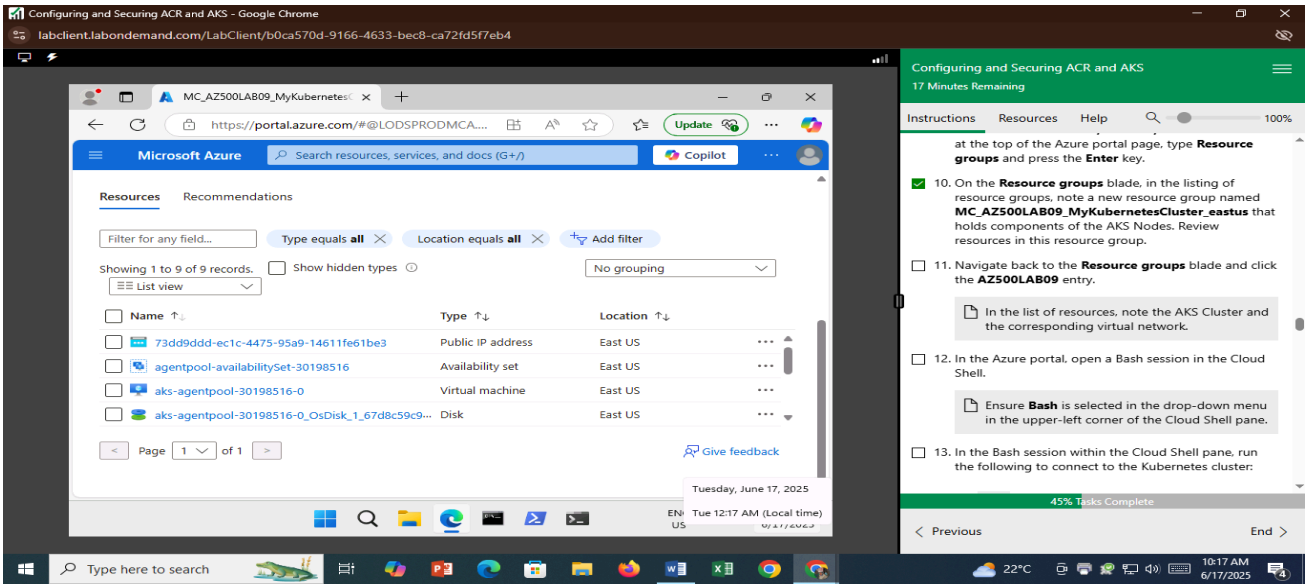


Step 9

In the Search resources, services, and docs text box at the top of the Azure portal page, I typed Resource groups and pressed the Enter key.

Step 10

On the Resource groups blade, in the listing of resource groups, I noted the new resource group named MC_AZ500LAB09_MyKubernetesCluster_eastus that holds components of the AKS Nodes and reviewed resources in that resource group.Here is my screenshot



Step 11

I navigated back to the Resource groups blade and clicked the AZ500LAB09 entry.

Step 12

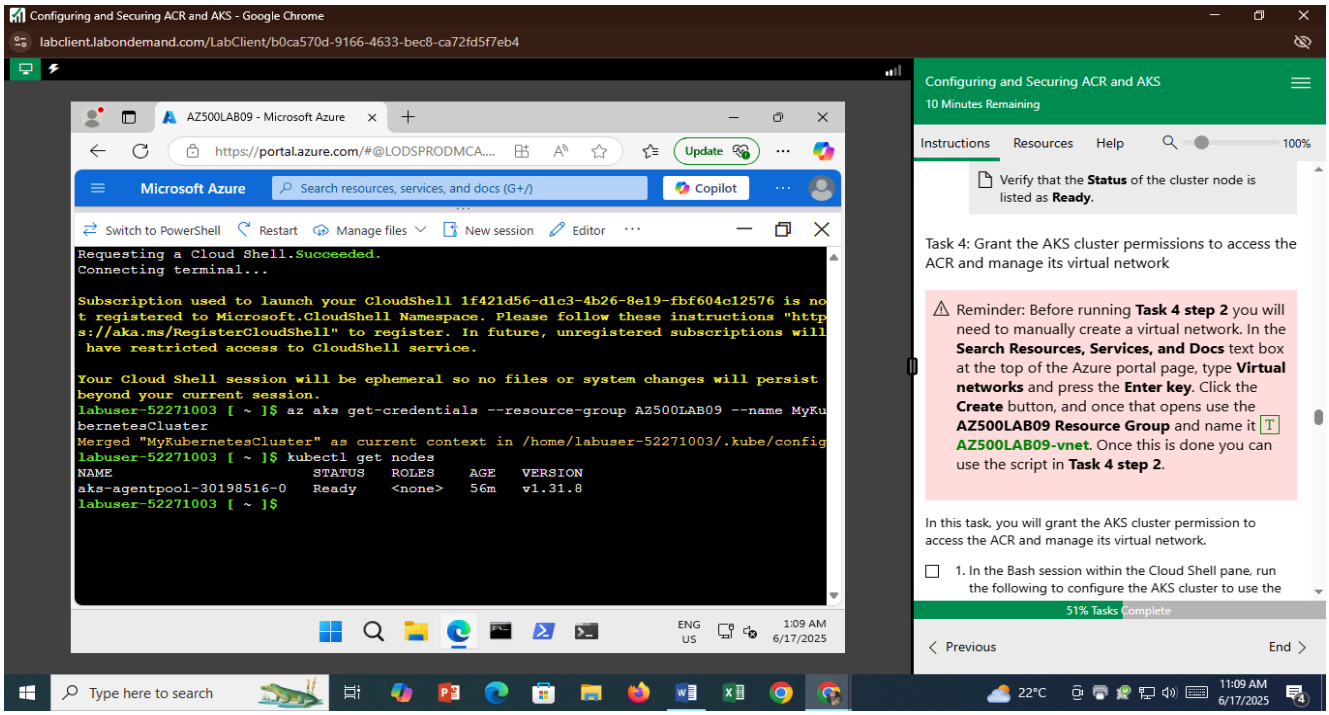
In the Azure portal, I opened a Bash session in the Cloud Shell.

Step 13

In the Bash session within the Cloud Shell pane, I ran the command to connect to the Kubernetes cluster.

Step 14

In the Bash session within the Cloud Shell pane, I ran the command provided to list nodes of the Kubenetes cluster and verified that the status of the cluster node is listed as Ready.Below is my screenshot

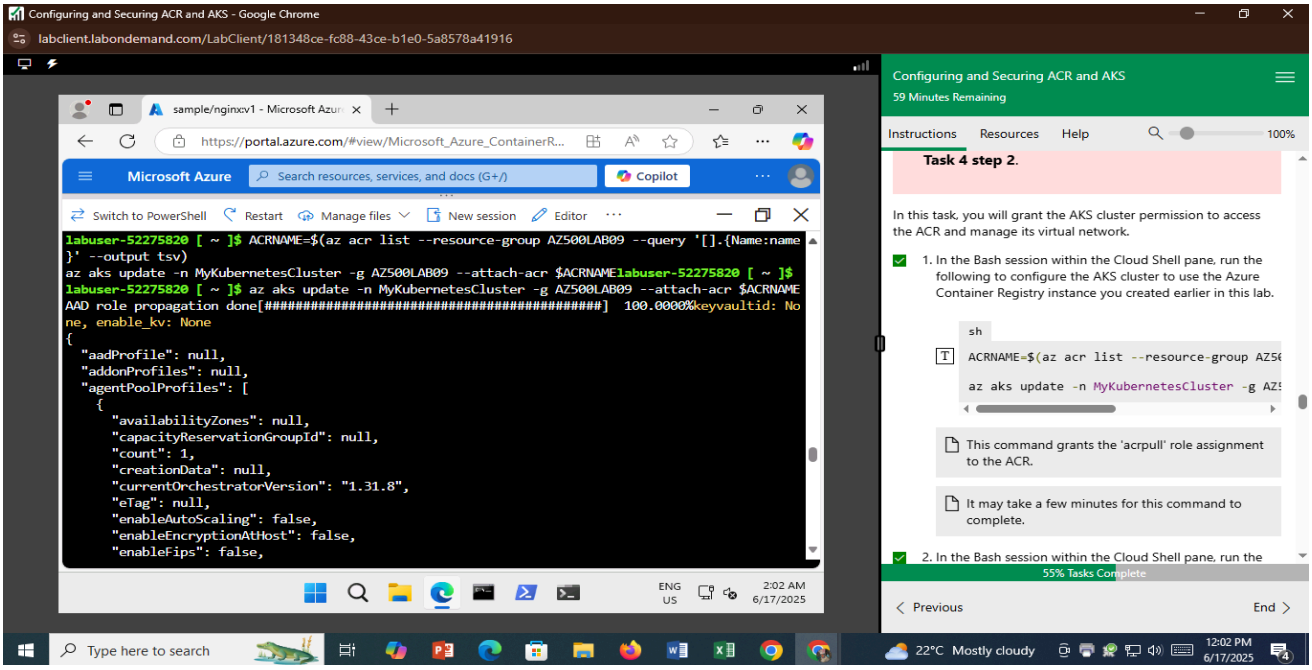


TASK 4

In this task, I granted the AKS cluster permission to access the ACR and manage its virtual network.

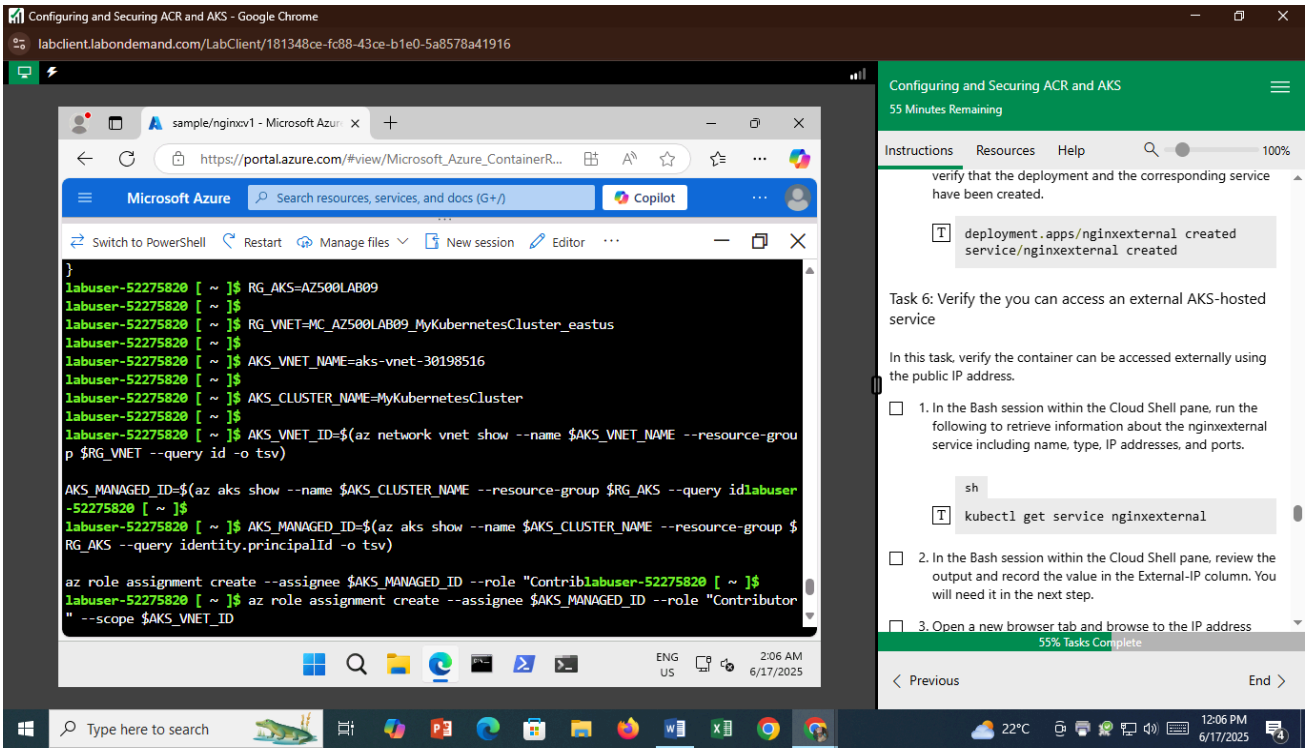
Step 1

In the Bash session within the Cloud Shell pane, I ran the command to configure the AKS cluster to use the Azure Container Registry instance I created earlier in the lab. The command granted the 'acrpull' role assignment to the ACR.Here is my screenshot



Step 2

In the Bash session within the Cloud Shell pane, I ran the command to grant the AKS cluster the Contributor role to its virtual network.Here is my screenshot



TASK 5

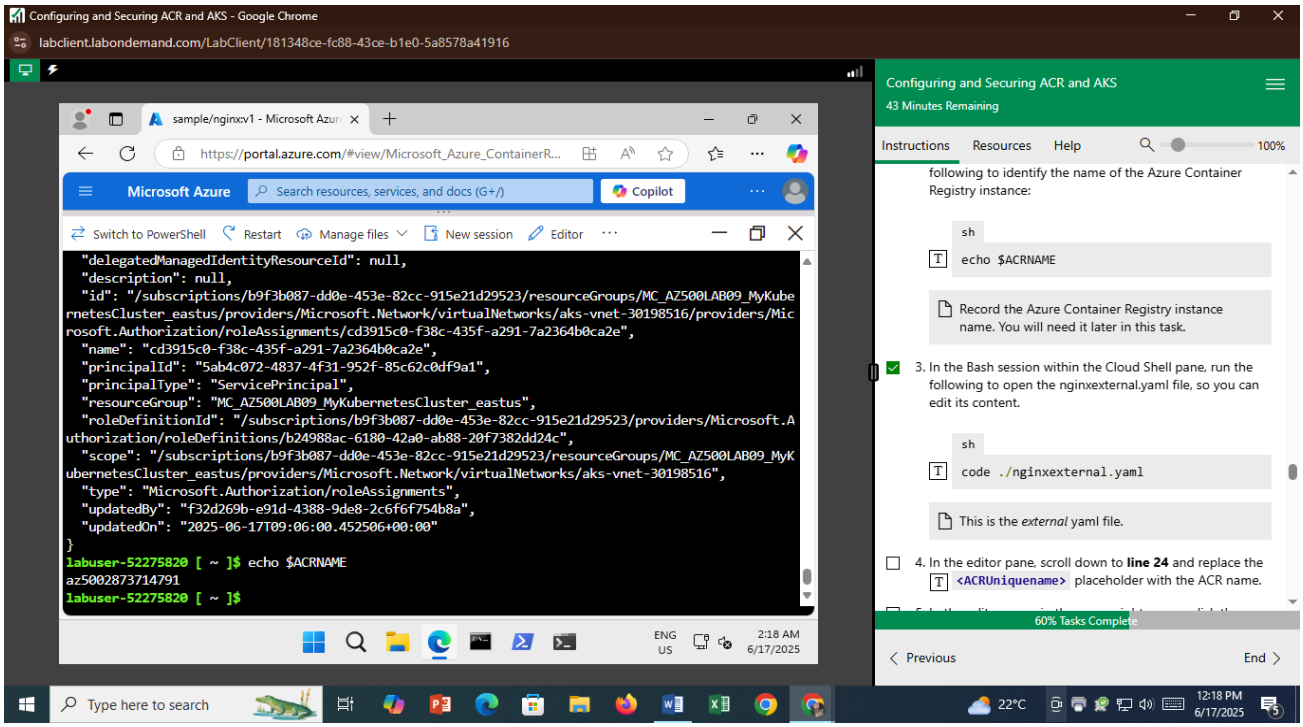
In this task, I downloaded the Manifest files, edited the YAML file, and applied my changes to the cluster.

Step 1

In the Bash session within the Cloud Shell pane, I clicked the Upload/Download files icon, in the drop-down menu, clicked Upload, in the Open dialog box, navigated to the location where I downloaded the lab files, selected \Allfiles\Labs\09\nginxexternal.yaml clicked Open. Next, selected \Allfiles\Labs\09\nginxinternal.yaml, and clicked Open.

Step 2

In the Bash session within the Cloud Shell pane, I ran the command provided to identify the name of the Azure Container Registry instance.Below is my screenshot

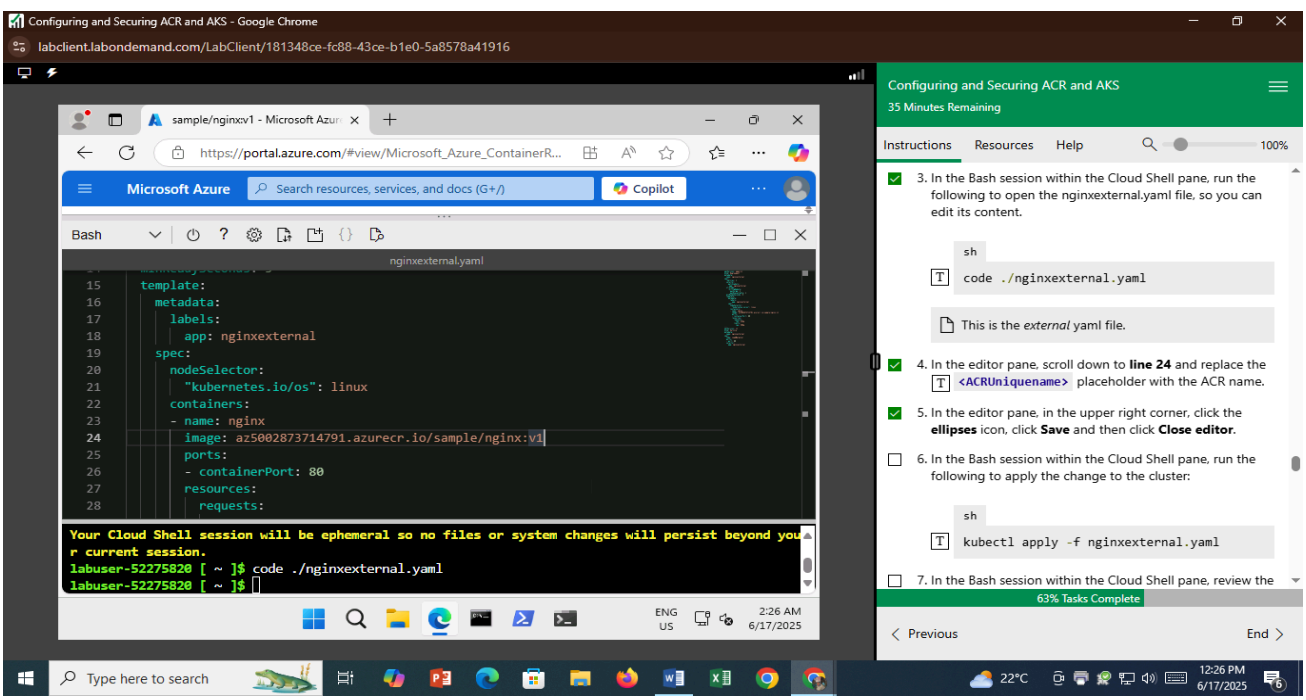


Step 3

In the Bash session within the Cloud Shell pane, I ran the command; code ./nginxexternal.yaml to open the nginxexternal.yaml file, so I can edit its content.

Step 4

In the editor pane, I scrolled down to line 24 and replaced the <ACRUniquename> placeholder with the ACR name.Here is my screenshot

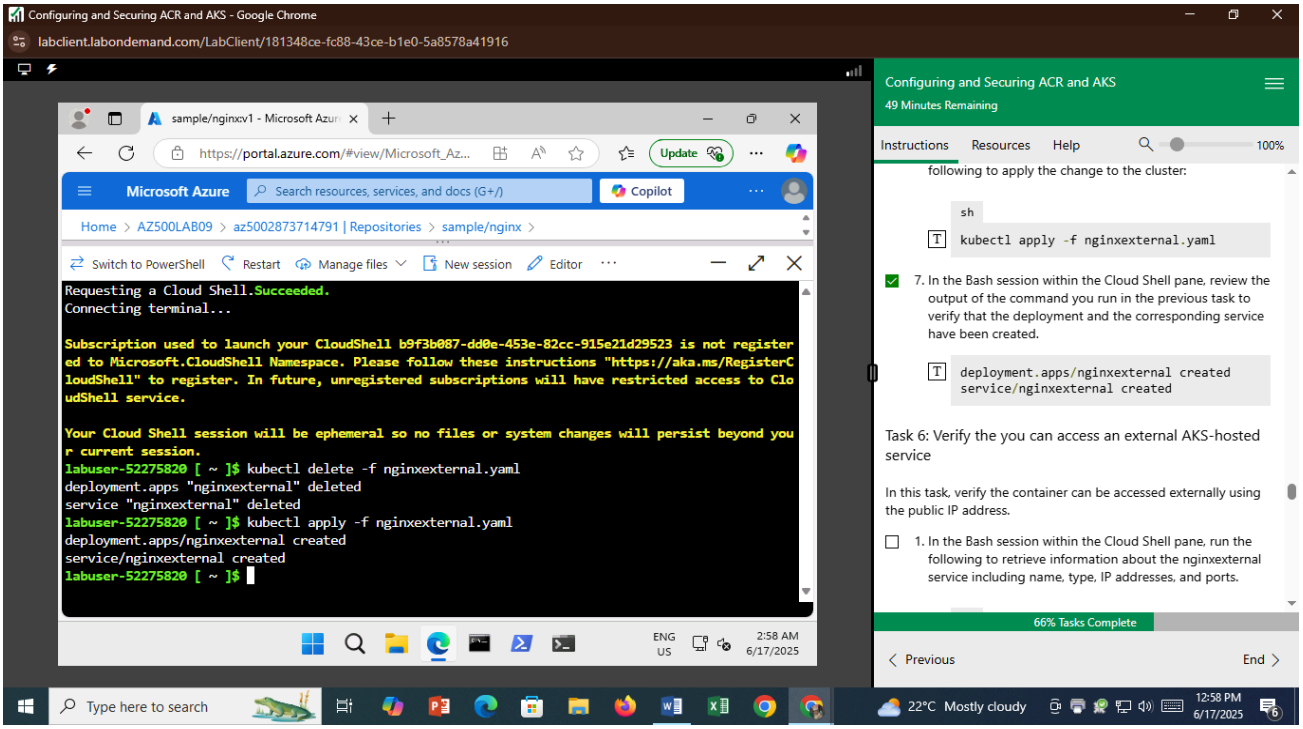


Step 5

In the editor pane, in the upper right corner, I clicked the ellipses icon, clicked Save and then Closed the editor

Step 6

In the Bash session within the Cloud Shell pane,I ran the command provided to apply the change to the cluster.Here is my screenshot



Step 7

In the Bash session within the Cloud Shell pane,I reviewed the output of the command ran in the previous step to verify that the deployment and the corresponding service have been created

TASK 6

In this task,I verified the container can be accessed externally using the public IP address.

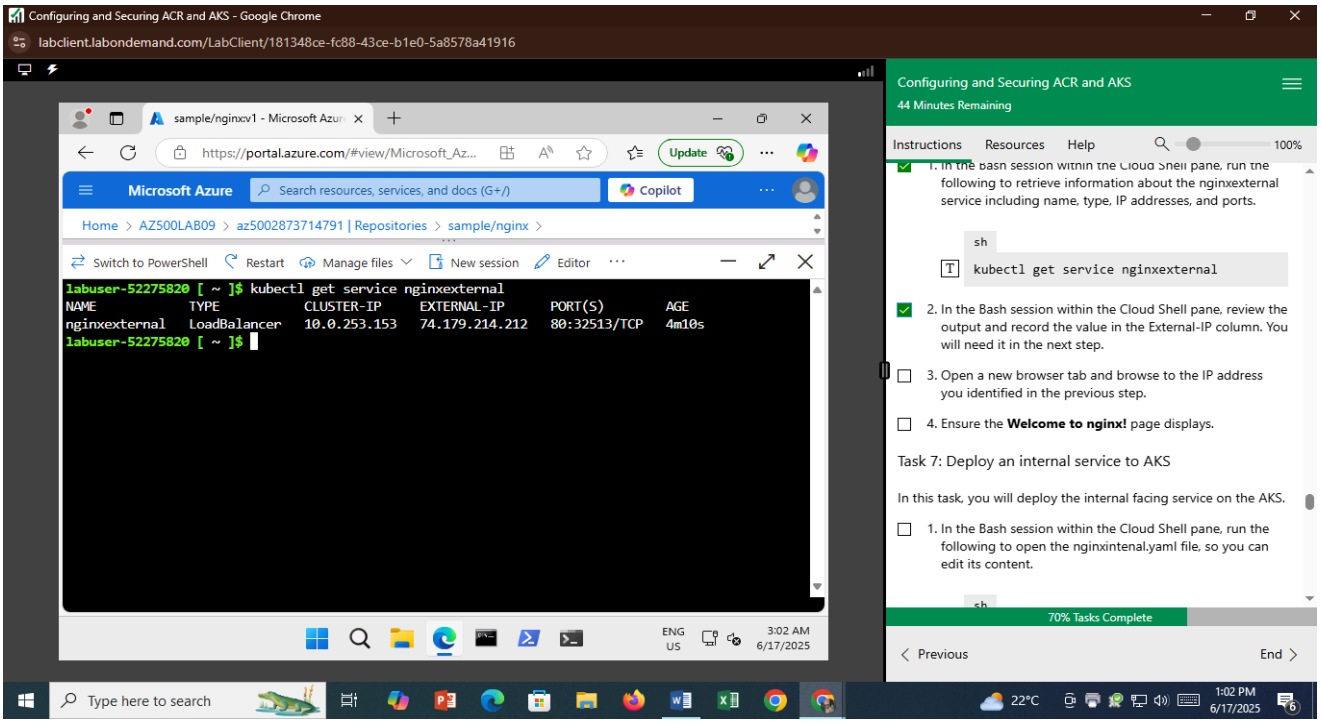
Step 1

In the Bash session within the Cloud Shell pane, I ran the command provided to retrieve information about the nginxexternal service including name, type, IP addresses, and ports.

Step 2

In the Bash session within the Cloud Shell pane, I reviewed the output and recorded the value in the External-IP column to use for the next step

Here is my screenshot for Step 1

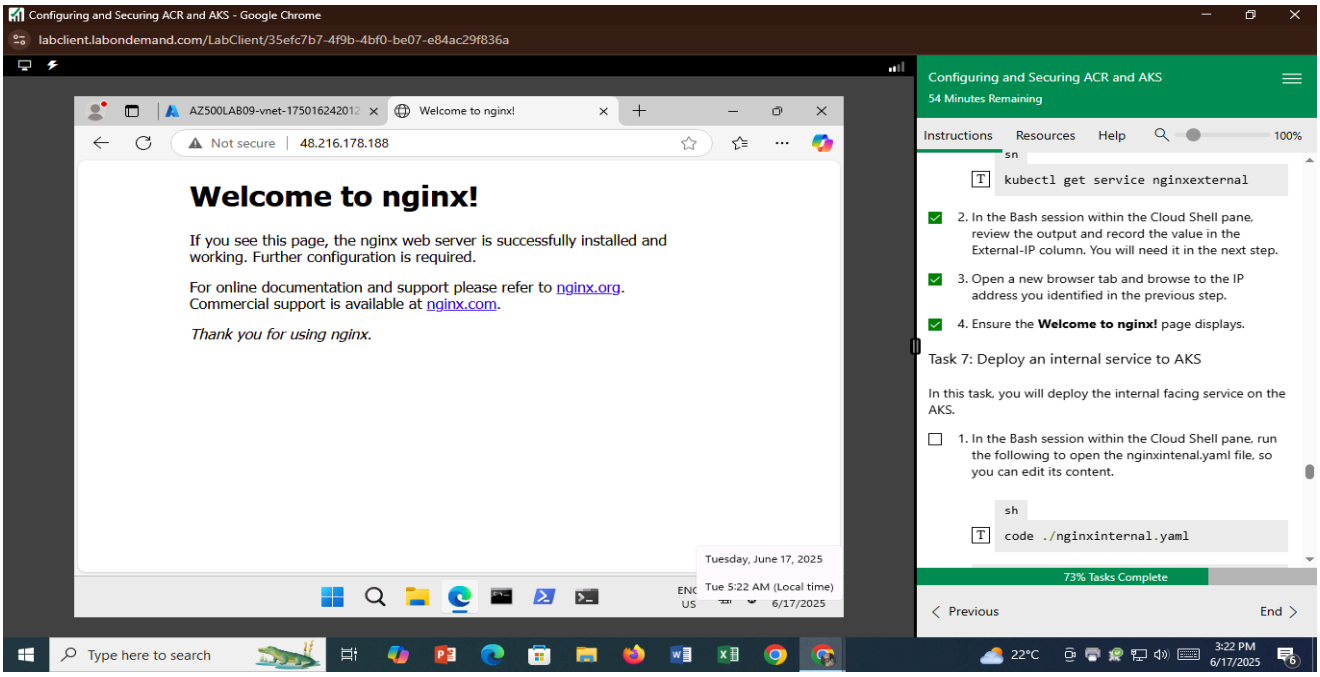


Step 3

I opened a new browser tab and browsed to the IP address I identified in the previous step

Step 4

Welcome to nginx! page displays.Here is my screenshot



TASK 7

In this task, I deployed the internal facing service on the AKS.

Step 1

In the Bash session within the Cloud Shell pane, I ran the command; code ./nginxinternal.yaml to open the nginxintenal.yaml file, so I can edit its content.

Step 2

In the editor pane,I scrolled down to the line containing the reference to the container image and replaced the <ACRUniquename> placeholder with the ACR name.

Step 3

In the editor pane, in the upper right corner, I clicked the ellipses icon, clicked Save and then closed the editor.

Step 4

In the Bash session within the Cloud Shell pane, I ran the command provided to apply the change to the cluster

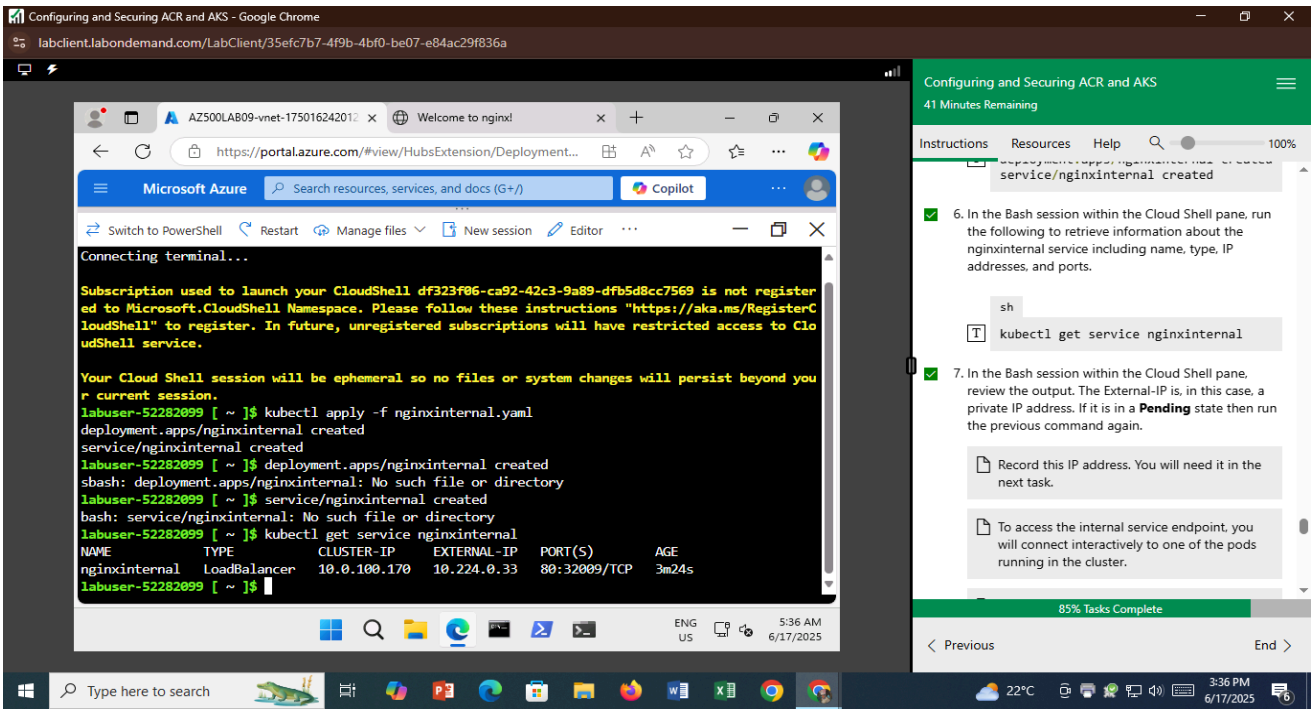
Step 5

In the Bash session within the Cloud Shell pane, I reviewed the output to verify my deployment and the service have been created

Step 6

In the Bash session within the Cloud Shell pane, I ran the command to retrieve information about the nginxinternal service including name, type, IP addresses, and ports.

Here is a screenshot for steps 4,5 and 6



Step 7

In the Bash session within the Cloud Shell pane,I reviewed the output. The External-IP is, in this case, a private IP address.I recorded this IP address as I would need it in the next task.

TASK 8

In this task, I used one of the pods running on the AKS cluster to access the internal service.

Step 1

In the Bash session within the Cloud Shell pane,I ran the command provided to list the pods in the default namespace on the AKS cluster.Here is my screenshot

Step 2

In the listing of the pods, I copied the first entry in the NAME column.

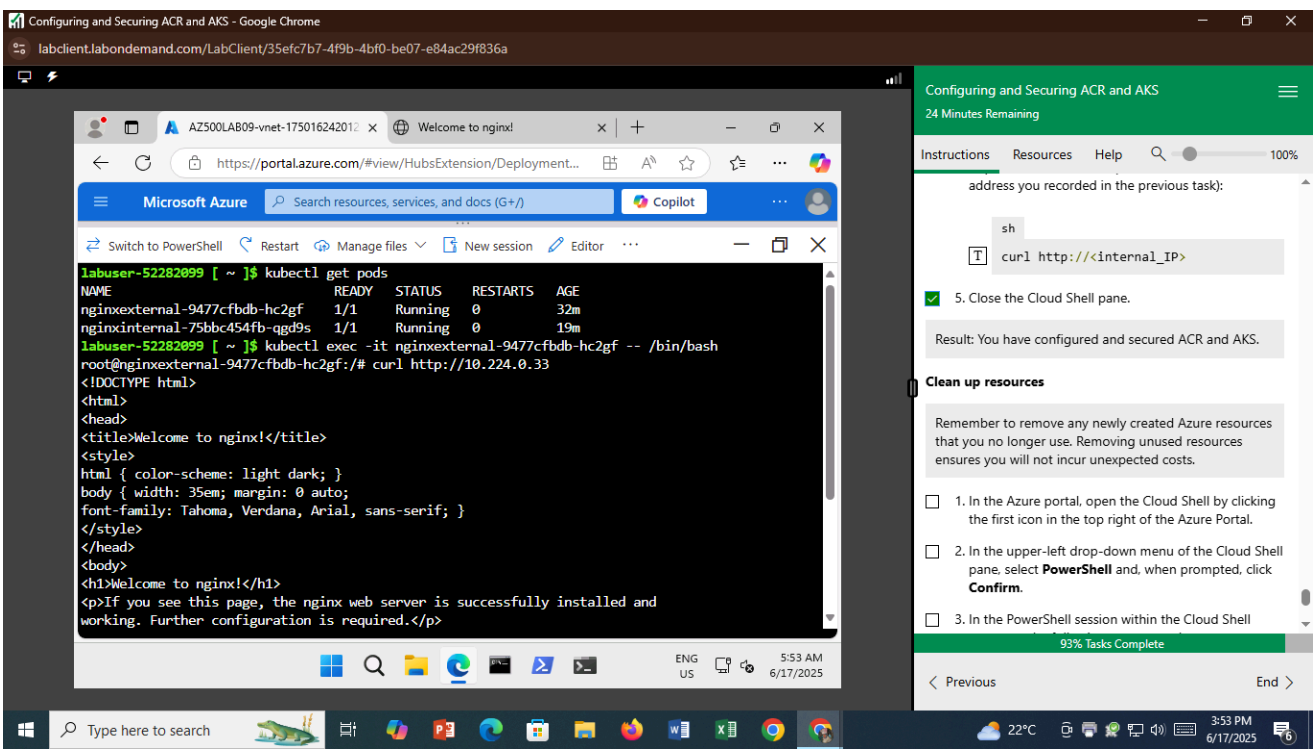
Step 3

In the Bash session within the Cloud Shell pane, I ran the command provided to connect interactively to the first pod placeholder with the name I copied in the previous step

Step 4

In the Bash session within the Cloud Shell pane,I ran the command provided to verify that the nginx web site is available via the private IP address of the service

Here is my screenshot for this task



Step 5

I closed the Cloud Shell pane,cleaned my resources and completed the lab